CAMERON WILDING

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Professional Summary

Electrical and Computer Engineering graduate student specializing in digital hardware design and hardware security. Experienced leading multi-person tape-out teams, developing and verifying RTL in Verilog/SystemVerilog, and PCB design. Seeking full-time hardware design beginning January 2027.

Education

Worcester Polytechnic Institute, Worcester, MA

Master of Science in Electrical and Computer Engineering, **GPA: 4.0/4.0** *May 2025 - Dec 2026*
Relevant Coursework: Computer Architecture, Advanced Digital Design

Bachelor of Science in Electrical and Computer Engineering, **GPA: 3.70/4.0** *Aug 2022 - May 2026*
Relevant Coursework: Advanced Digital Design with FPGAs, Computer Organization, Analog IC Design, CMOS Fundamentals

Publications

Wilding, C., Shaker, M., et al. “Not to Break, but to Attest: Adversarial Probes for Privacy-Preserving LLM Verification”. Under review. Preprint: https://arxiv.org/abs/2608.27954

Work Experience

Graduate Research Assistant, Worcester Polytechnic Institute *Jan 2026 – Present*

- Conducted research on cryptographic AI governance by developing a system to verify large language model integrity without access to model weights or internals using adversarial probes, as part of M.S. thesis
- Analyzed RF emissions from GPUs running AI workloads to extract model information and generated tokens via cache-clock side-channel analysis

Teaching Assistant, Worcester Polytechnic Institute *Aug 2025 – Jan 2026*

- Supported 31 students in ECE3829: Advanced Digital Design with FPGAs
- Guided development of digital circuits and state machines on the Xilinx Artix-7 in Vivado
- Graded lab work and provided technical support in Verilog

Projects

Protocol Emulator ASIC, Jane Street ASIC Design Competition *Oct 2026 – Present*

- Designing a programmable core that implements UART, SPI, I2C, and other protocols in firmware rather than fixed logic
- Targeting IHP 130nm CMOS5L via Tiny Tapeout under a 0.7 mm^2 area budget, verifying with cocotb testbench

Electroencephalography ASIC, IEEE Solid-State Circuit Society *Sep 2026 – Present*

- Lead digital design developing a custom mixed-signal EEG ASIC using the Cadence tool suite
- Architecting memory controller and custom 8-bit Digital Signal Processing focused CPU optimized for low on-chip cache
- Work with EE undergraduates to educate and guide chip design

Power Management IC Design for IoT Applications, WPI Major Qualifying Project *Aug 2025 – May 2026*

- Led digital design for an 8-person team developing a custom ASIC using the Cadence tool suite
- Designed control state machines in Verilog to manage power monitoring and sequencing
- Won second place among all WPI Major Qualifying Projects; taped out May 2026, with testing planned for Fall 2026

Trace Data Compressor, Independent *April 2026*

- Designed a lossless data compressor in SystemVerilog to compress bus data for storage on BRAM
- Achieved 7.4x compression on FPGA, winning Best Rookie Hack at GoatHacks 2026; taped out via TinyTapeout 130nm

Tensor Co-Processor for Croc SoC, Independent *Oct 2025 – Dec 2025*

- Designed systolic array, control logic, and partial product accumulation in SystemVerilog, optimized for low area/power
- Verified with UVM in December 2025; taped out on TSMC 180NM in April 2026, with testing planned for Fall 2026

Technical Skills

Hardware/Software: Vivado, KiCad, LTspice, Altium, Git, Cadence Tool Suite (Virtuoso, Genus, Innovus, Tempus, Spectre), SPICE, Linux, UVM, PyTorch, FPGA, LaTeX

Programming Languages: SystemVerilog, Verilog, C, Python, MATLAB